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Section $\rightarrow$ 4 (CSE)

Course Code $\rightarrow$ MAS2001

Assignment $\rightarrow$ 4

Date

Section - A

- Q1)
- (i) Statistical Hypothesis $\Rightarrow$ It is a statement or assumption about a population parameter that can be tested using sample data.
 - (ii) Null hypothesis $\Rightarrow$ It is the default assumption that there is no effect or no difference.
Alternate Hypothesis $\Rightarrow$ It is the statement we are trying to find evidence for, suggesting there is an effect or a difference.
 - (iii) One-tailed test $\Rightarrow$ A test that looks for a diff. in one specific direction.
Two-tailed test $\Rightarrow$ A test that looks for any diff., regardless of direction.
 - (iv) Type I error $\Rightarrow$ Rejecting a true null hypothesis.
Type II error $\Rightarrow$ Failing to reject a false null hypothesis.
 - (v) Significant level $\Rightarrow$ The threshold probability (alpha) for deciding whether to reject the null hypothesis.
Critical Region $\Rightarrow$ The set of values for the test statistic that leads to rejection of the null hypothesis.

Date

(vi) Confidence Interval $\Rightarrow$ A range of values of an unknown parameter, calculated from the sample data.
Confidence Limit: The upper and lower bounds of the confidence level.

- Q2) Null hypothesis
- Q3) Type I and Type II errors
- Q4) alpha (α)
- Q5) I, α
- Q6) Rejection
- Q7) (b) Accepting an incorrect hypothesis
- Q8) (d) accepts a null hypothesis when it is false
- Q9) (c) rejects it, when it is true

Section - B

- Q10) 10 adult, $\mu = 152$, $\sigma^2 = 32.0625$
 $H_0 =$ the population mean height = 162.5 cm
 $\mu = 162.5$ cm
 $H_1 \Rightarrow \mu < 162.5$ cm

Thus one is one tailed test bcz the alternative hypothesis is directional

(B) avg length = 8 mm

sample = 18 cells
mean = 7.2 mm
st dev = 0.5 mm
 $\alpha = 0.05$

This is two tailed test because the alternative hypothesis does not specify a direction

$\therefore$ go for t-test

$H_0 \Rightarrow \mu = 8$ mm
 $H_1 \Rightarrow \mu \neq 8$ mm

$$t = \frac{7.2 - 8}{0.5/\sqrt{18}} = -1.7$$

$\alpha = 0.05$, $df = n-1 = 17$
critical = ± 2.11

-1.7 lies within $-2.11 < t < 2.11$
 $\therefore$ does not fall in rejection region

at $\alpha = 0.05$ significant level, we do not have sufficient evidence to reject the null hypothesis

Q11) (A)

f-test $F = \frac{S_1^2}{S_2^2}$ if $S_1^2 > S_2^2$ else $F = \frac{S_2^2}{S_1^2}$
 $S_1^2 = (0.75)^2 = 0.5625$, $S_2^2 = (0.25)^2 = 0.0625$
 $F = \frac{S_1^2}{S_2^2} = \frac{0.5625}{0.0625} = 9$

$df_1 = n_1 - 1 = 13 - 1 = 12$
 $df_2 = n_2 - 1 = 12 - 1 = 11$

for two tailed f-test at $\alpha = 0.05$

$df_1 = 12$ and $df_2 = 11$

critical lower = 0.39, critical upper = 2.93

$\therefore F = 9$, it lies below range $0.39 < F < 2.93$

if $\alpha = 0.05$ significant, we do not reject the null hypothesis

(B)

$\bar{x} = \frac{8200 + 8130 + 8310 + 8070 + 12340}{5}$
 $\bar{x} = 8320$

variance = $\frac{\sum (x_i - \bar{x})^2}{n-1}$
 $= 15750$

Min-2, $s_2 = \frac{7950 + 7850 + 7900 + 7920 + 7840}{5}$
 $= 7956$

Date

$$\text{variance} = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$= 11256.07$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$= \frac{8230 - 7956.17}{\sqrt{\frac{15700}{5} + \frac{11256.07}{6}}}$$

$$t = \frac{273.33}{\sqrt{3150 + 1876.02}} = \frac{273.33}{70.91} = 3.85$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}} = 16$$

for $\alpha = 0.05$, $df = 16$, two tailed critical
t-value
critical = 1.9306

$\therefore t = 3.85 > 1.9306$, we reject H_0

Q12) $\bar{x} = 1002$ $t = \frac{1002 - 1000}{14/\sqrt{10}}$
 $\mu = 1000$
 $s = 14$ $t = 0.453$
 $n = 10$

Date

one tailed test $\alpha = 0.05$, $df = n-1 = 9$
 $t_{critical} = 1.833$

$t = 0.453$ & $t_{critical} = 1.833$ does not fall
in the rejection region

At the $\alpha = 0.05$, we do not reject the null
hypothesis

(B)

 $n = 10$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$\bar{x}$ = Sum of all values / no. of vals

$$\bar{x} = \frac{50 + 40 + 44 + 32 + 45 + 48 + 40 + 45 + 40 + 43}{10}$$

$$= 42.3$$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$= \frac{(50-42.3)^2 + \dots + (43-42.3)^2}{9}$$

$$= \frac{72.1}{9} = 8.01$$

$$t = \frac{42.3 - 50}{\sqrt{8.01}/\sqrt{10}}$$

$$= -3.02$$

$$t_{critical} = \pm 2.662$$

$t = -3.02 < -2.662$ $\therefore$ falls in rejection
region

At $\alpha = 0.05$, we reject null hypothesis

Date

Section C

Q13) $f(n,u) = 2$ for $0 \leq n \leq 8$

(i) Critical region $0.5 < n$

Type I error (α):

$$\alpha = P(\text{reject } H_0 | H_0 \text{ is true}) = P(n > 0.5 | 0.1)$$

$\therefore n$ is uniformly distributed on $[0,1]$
under H_0 :

$$\alpha = 1 - 0.5 = 0.5$$

Type II error (β):

$$\beta = P(\text{accept } H_0 | H_1 \text{ is true})$$

$$= P(n \leq 0.5 | 0.5)$$

Since n is uniformly distributed on $(0,1)$
under H_1 :

$$\beta = 0.5/1 = 0.5$$

(ii) Critical region $1 \leq n \leq 1.5$

Type I error (α):

$$\alpha = P(\text{reject } H_0 | H_0 \text{ is true}) = P(1 \leq n \leq 1.5 | 0.1)$$

Since n can't exceed 1 under H_0 .

$$\alpha = 0$$

Type II error (β):

$$\beta = P(\text{accept } H_0 | H_1 \text{ is true}) = P(n < 1 \text{ or } n > 1.5 | 0.5)$$

Date

under H_0 , n is uniformly distributed on $(0,1)$:

$$P(n < 1) = 1/1, \quad P(n > 1/5) = \frac{2 \times 1/5}{2} = 0.25$$

$$\therefore \beta = 0.5 + 0.25 = 0.75$$

Q14)

$$H_0: p = 1/2$$

$$H_1: p = 3/4$$

we need to calculate $P(n > 2)$ when $p = 1/2$

$$P(n=4) = 5 \times \left(\frac{1}{2}\right)^4 \times \left(\frac{1}{2}\right)^1 = 5 \times \frac{1}{8} \times \frac{1}{2} = 5/32$$

$$P(X=5) = 1 \times \left(\frac{1}{2}\right)^5 = 1/32$$

$$\text{Total prob} = P(n=4) + P(n=5)$$

$$= 5/32 + 1/32 = 6/32 = 3/16$$

Q15)

$$H_0: \theta = 2$$

$$H_1: \theta = 1$$

Critical region $n \geq 1$

Type I error $\alpha = P(n \geq 1 | \theta = 2)$

$$f(n,u) = 9e^{-9n} \text{ for } \theta = 2$$

$$f(n,1) = 2e^{-2n}$$

$$\alpha = \sum_{n=1}^{\infty} 2e^{-2n}$$

Date

$$\alpha = 2 \left[\frac{1 - e^{-2}}{2} \right]$$

$$\alpha = 2 \left(0 - \left(-\frac{e^{-2}}{2} \right) \right)$$

$$\alpha = e^{-2}$$

Type II error

$$\beta = P(n < 1 | \theta = 1)$$

$$f(n+1) = e^{-n}$$

$$P_n = \int_0^1 e^{-n} du$$

$$P_n = [-e^{-n}]_0^1$$

$$\beta = (-e^{-1}) - (-1)$$

$$\beta = 1 - e^{-1}$$

$$\beta = 1 - 1/e$$

(10) critical region = $n \geq 6$ (reject H_0 and accept H_1)

Acceptance region $n < 6$ (Accept H_0)

Type I error (α):

$$\alpha = P(n \geq 6 | \theta = 2) = 1 - P(n < 6 | \theta = 2)$$

$$P(n < 6 | \theta = 2) = 1 - e^{-6/2} = 1 - e^{-3}$$

$$\therefore \alpha = e^{-3} = 0.0498$$

Date Type II error (β):

$$\beta = P(n < 6 | \theta = 4)$$

$$P(n < 6 | \theta = 4) = 1 - e^{-6/4} = 1 - e^{-1.5}$$

$$\beta = 1 - e^{-1.5} \approx 1 - 0.2231$$

$$\beta = 0.7769$$